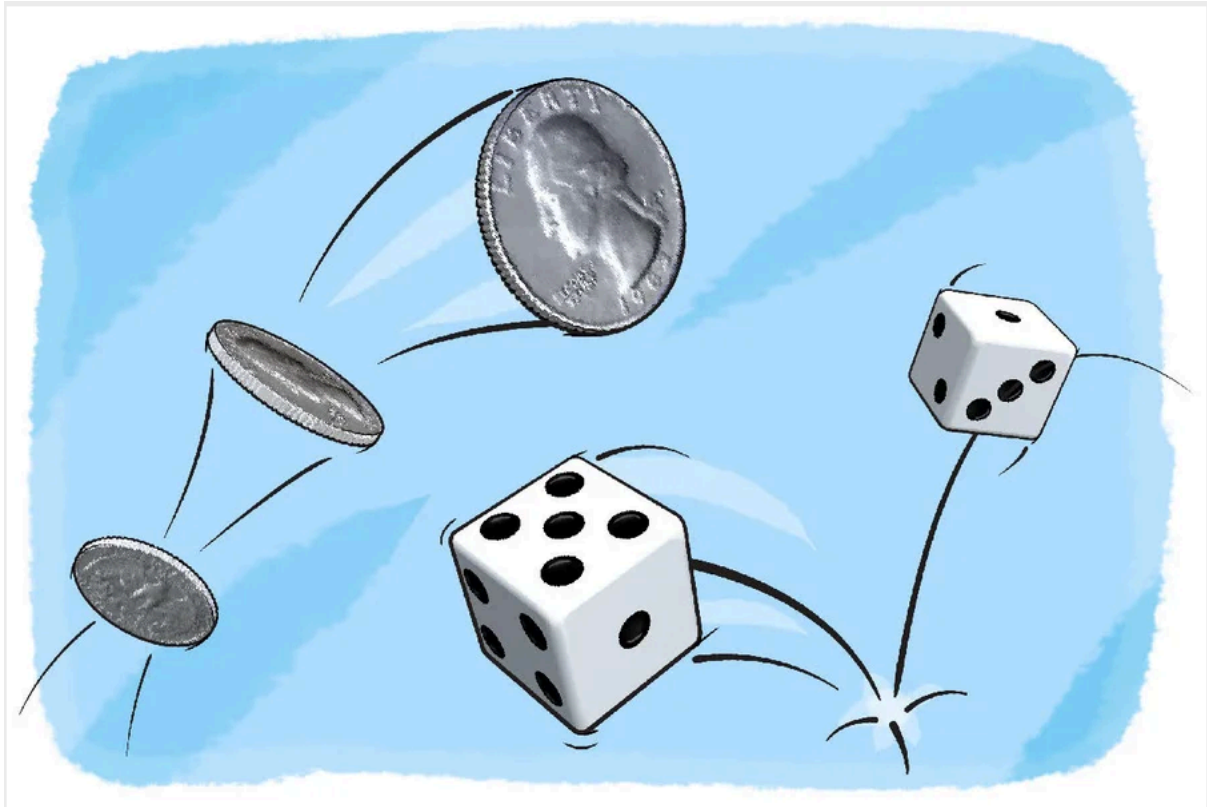




Probability 2024

Module Leader: Diomides Mavroyiannis
Contact: Inquiries during the teaching term should be sent via Canvas messages. Before and after the term, students can reach the module leader through the Academic Team (academics@msinst.org).

1. Overview

1.1. Short Description

The module covers major topics of probability theory and statistics. We will introduce theoretical concepts through simple experiments and games. Then we discuss some important probability distributions: what kind of distribution we should use for lifetimes/waiting times, or for the number of mistyping in the pages of a newspaper, etc. How can we describe a probability distribution: distribution, density functions (and also characteristic functions)? Law of large numbers in a mathematical formulation. Central limit theorem - or why the

Gaussian (Normal) distribution is special. These concepts are used in financial mathematics. The above concepts form the basis of statistics: giving tools for appraising the world around us. Estimates, hypothesis testing. These are used in social sciences and medicine as well. The module is highly recommended for those wishing to study Mathematics, Economics, or other numerical sciences at university, but students interested in some fields of natural sciences are also welcome.

1.2. Module Format and Credits

Level: Immersion 2

Division: Numerical Sciences

Pathway: Mathematics and Its Applications

Module Format: eight 50-minute sessions over 8 consecutive weeks.

1.3. Learning outcomes

Students will understand the scope and use of probability in mathematics and the fundamentals of this field. They will improve their problem-solving abilities with standard techniques related to probability applied to various problems.

1.4. Prerequisites

The differentiation module is a recommended prerequisite - basic knowledge of differentiation is essential. Knowledge integration is helpful but not required. It is suggested for students who are more interested in mathematics or the mathematical aspects of economics. Basic knowledge of differentiation: one can get all that is needed by watching Gilbert Strang's [series](#) for high school students.

1.5. Module Preparation

You are advised to read Sections 1-4 from Chapter 1 in the Ross book required below on basic counting techniques. You can also self-test yourself by trying exercises on p20.

For those who did not take *Integration* or do not know integrals, **before the 5th session**, watch the set of videos and read the notes found at the following links (and their left-side menu)

- <https://www.khanacademy.org/math/calculus-1/cs1-integrals/cs1-accumulations-of-change-introduction/v/introduction-to-integral-calculus>
- [Links to an external site.](#)
- <https://www.khanacademy.org/math/calculus-1/cs1-integrals/cs1-fundamental-theorem-of-calculus-and-definite-integrals/v/connecting-the-first-and-second-fundamental-theorems-of-calculus>

The book we use is [Sheldon Ross, A First Course in Probability](#)

[Links to an external site.](#),

Another book that I recommend for those enthusiastic about the topic is

[Blitzstein and Hwang, Introduction to Probability](#)

2. Module Structure

2.1. Week-by-Week Lesson Plan

2.1.1. Session 1 - *Events, Probability Space, Classical Probability, Conditional Probability.*

Session Summary

The notion of an event is introduced, as well as those of probability and their basic rules and properties. The probability model is introduced. The important idea of conditional probability is introduced.

Required Reading: Ross. Ch. 2-3

Optional Reading: Blitzstein and Hwang, Ch. 1-2

2.1.2. Session 2 - More Conditional Probability, The Law of Total Probability, Bayes' Theorem, Stochastic Independence

Session Summary

We cover the law of total probability and apply it to some applications using the probability tree diagram method. We consider Bayes' theorem a fundamental

theorem on conditional probability. We also take the notion of stochastic independence, one of the central notions of probability theory.

Required Reading: Ross, Ch 2-3

Optional Reading: Blitzstein and Hwang, Ch. 1-2

2.1.3. Session 3 - *Random Variables, Cumulative Distribution Function. Discrete Random Variables. The Probability Mass Function. Expectation and Variance in the Discrete Case.*

Session Summary

We introduce the notion of a random variable. We consider their laws, such as the cumulative distribution function (CDF) and its properties. We take the probability mass function (PMF), a substitute for the CDF in the discrete case. We also introduce discrete random variables' expected value and variance and state their elementary properties.

Required Reading: Ross, Ch 4.1-4.5

Optional Reading: Blitzstein and Hwang, Ch. 3-4

2.1.4. Session 4 - Important Discrete Random Variables

Session Summary

We review the list of well-known discrete random variables: Bernoulli, Binomial, Poisson, Geometric (or Pascal), Hypergeometric, Negative Binomial, and their parameters with intuitive explanations.

Required Reading: Ross, 4.6-4.8

Optional Reading: Blitzstein and Hwang, Ch. 3-4

2.1.5. Session 5 - *Continuous Random Variables. Probability Density Function. Expectation and Variance in the Continuous Case*

Session Summary

The second important class of continuous random variables is introduced. We take the probability density function (PDF), a substitute for the CDF in the constant case. We also introduce the expected value of a continuous random

variable, hence its variance. For those unfamiliar with integration, we will do a short recap here.

Required Reading: Ross, 5.1-5.2

Optional Reading: Blitzstein and Hwang, Ch. 5

2.1.6. Session 6 - Important Continuous Random Variables

Session Summary

We review the most important continuous random variables with Uniform, Exponential, and Normal distributions. We also review their relationship to their discrete counterparts. We discuss the critical parameters of these distributions mainly informally.

Required Reading: Ross, 5.3-5.5

Optional Reading: Blitzstein and Hwang, Ch. 5

2.1.7. Session 7 - The Law of Large Numbers and the Central Limit Theorem

Session Summary

The Law of Large Numbers and perhaps the most beautiful and important theorem of mathematics the Central Limit Theorem. We illustrate the theorem and show how it can be applied to problems otherwise intractable. We also argue for the importance of normal distribution in probability theory and mathematical statistics.

Required Reading: 8.1-8.3

Optional Reading: Blitzstein and Hwang, Ch. 10

2.1.8. Session 8 - Mathematical Statistics

We consider the basic point of view of statistics and study the notions of confidence interval and hypothesis test.

Required Reading: [Keijo Ruohonen: Statistics 1](#)

[Links to an external site.](#), Ch. 2

Optional Reading: [Keijo Ruohonen: Statistics 1](#)

[Links to an external site.](#), Ch. 3

3. Assessment

3.1. Assessment Types and Weighing

- Weekly assignments (50%): Some exercises on the current class material.
- Longer set of problems (30%): there will be one after class 4, and one after class 8 carrying equal weight
- Exam (20%): A written exam the week after the term's last class from the term's material.

3.2. Requirements

To receive full marks, students must explain **every step of the solution** (both calculations and explanatory texts) cleanly and with a well-developed structure. Usually, no mark is given for declaring a (possibly correct) final result without verification.

Plagiarising or copying a fellow student's work results in zero points for all parties involved. Students should understand every part of their submission and need to be able to explain their solution to the module leader if requested.

All submissions are online by uploading a single pdf file. You can write your solutions on paper, scan and convert them to PDF, or type them using LaTeX and upload the generated PDF.

If you cannot attend a class for a valid reason, you must inform the **module leader before class**. Valid absences are, for example, physical or psychological problems validated by a doctor or guardian and highly significant academic activities validated by a mentor.

3.3. Deadlines

Submission deadlines are indicated on the Canvas course page. Late submissions are not accepted in the weekly assignments during the Term. In the case of the final assignment, every day of late submission results in a 10% deduction from the awarded grade. No submission is accepted more than three days after the final coursework submission date.

3.4. Grading

The grading scheme works on a 1-10 scale, with 1-3 being a fail and 8-10 being exceptional, for which extra work beyond the requirements is needed. Indicative grade boundaries are 95%, 85%, 75%, 65%, 55%, 45%, 35%, 25%, 15%.

3.5 Feedback

Feedback is provided as in-line commentary, annotations, and overall comments no later than the final grading deadline.

4. Academic Handbook

In any case not described in the Syllabus, the rules of the Academic Handbook apply.